

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Energy & material cycles: natural vs human systems - Set 1

Mark scheme

AQA spec 3.6.2-3.6.3; Sustainability; textbook pp. 373-374

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Natural processes are driven by renewable, low energy-density solar energy captured by photosynthesis; human processes usually use high energy-density non-renewable fossil fuels (1) • Natural processes occur at low temperatures because enzymes lower the activation energy of reactions; human processes need high temperatures and pressures (1) • Natural processes produce no toxic wastes / their products are raw materials for other processes; industrial processes emit CO₂ and NO_x (1) • So natural fixation is sustainable indefinitely with a low carbon footprint (1)	4
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Hydrological cycle (evaporation, precipitation) (1) • Carbon cycle / photosynthesis (1) • Atmospheric circulation / winds (1) • Thermohaline circulation (density differences from heating / evaporation) (1) • Decomposition (1)	3
01.3	<i>1 mark for each point.</i> • Bioleaching and phytomining extract metals at low temperature instead of high-temperature smelting (1) • Anaerobic digestion and composting treat wastes; enzymes in industry (biological washing powders, biofuels from algae / microorganisms) (1) • Heat pumps and solar thermal capture low-density heat; low-temperature manufacturing using biological polymers; fuel cells instead of combustion (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
02.1	<i>1 mark for each point.</i> - Natural cycles use a small number of abundant elements (C, H, O, N, P) built into monomers and polymers (carbohydrates, proteins); wastes of one process are raw materials for another so nothing is depleted or accumulates (1) - Natural wastes are biodegradable / non-toxic and are broken down by decomposers into raw materials (1) - Human systems are linear: extract - process - manufacture - use - discard; minerals are dispersed after use or mixed so recovery is difficult (1) - Human wastes are often persistent and toxic (organochlorines, PCBs, heavy metals in electronics) and build up; fossil fuel use is inherently linear and unsustainable (1)	4
02.2	<i>1 mark for each point.</i> - Metals are dispersed in small amounts within complex mixed products so they cannot easily be separated / recovered at end of life (1) - They leach from landfill or are released by incineration; persistent, bioaccumulating (1)	2
02.3	<i>1 mark for each point.</i> - Design for durability, repair and re-use (replaceable parts, no built-in batteries, spare parts and support, no planned obsolescence) (1) - Easy identification of materials (coding) and separation of components so they can be recycled (1) - Separation of technical materials (metals, plastics) from biological materials (which can be composted / digested) (1) - Use of recycled and recyclable / biodegradable materials; reduced packaging and material use (Cradle to Cradle) (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.3.	
03.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Hydrological cycle (1) - Carbon cycle (1) - Nitrogen cycle (1) - Atmospheric circulation (1) - Thermohaline circulation (1) - Photosynthesis (1)	2
03.2	<i>1 mark for each point.</i> - Enzymes reduce the activation energy of reactions (1) - So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures (1)	2
03.3	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> - Haber process (nitrogen fixation) (1) - Incineration of wastes (decomposition) (1) - Many manufacturing processes (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Energy: low energy-density renewable (solar) / non-renewable fossil fuels (1) • Materials: cyclic - wastes of one process are raw materials of the next / linear - used once then discarded, depletion (1) • Wastes: biodegradable, non-toxic or do not build up / toxic, persistent (eg PCBs, organochlorines, toxic metals) that bioaccumulate (1) • Temperature: low / high (Haber process, incineration) (1)	4
04.2	<i>1 mark for each point.</i> • Abundant simple raw materials build into monomers and then a wide range of polymers (carbohydrates, proteins) (1) • They are re-used in cycles so there is no depletion of raw materials or build-up of wastes (1)	2
04.3	<i>1 mark for each point.</i> • Use often disperses the minerals (eg in products spread over the world / corrosion) (1) • Or produces mixtures of materials that are difficult to separate, so sustainable exploitation is reduced (1)	2
04.4	<i>1 mark for each point.</i> • They cannot easily be removed from end-of-life appliances and are not biodegradable (1) • Natural wastes are non-toxic / broken down to raw materials for other processes, whereas persistent toxins may bioaccumulate and biomagnify (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373-374; AQA spec 3.6.2-3.6.3.	
05.1	<i>Correct answer only.</i> • Waste products that are the raw materials for other processes (1)	1
05.2	<i>1 mark for each point.</i> • Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) (1) • The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂ (1)	2
05.3	<i>1 mark for each point.</i> • Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials (1) • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	